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Quick release motor pump assembly for electric marine toilet

Abstract

Apparatus featuring a toilet base having a wall forming a chamber for receiving waste, the wall having a toilet base coupling portion; a pump/motor assembly having a pump inlet configured to receive the waste pumped from the chamber, the pump inlet having a pump inlet coupling portion arranged inside the toilet base coupling portion when the pump/motor assembly and the toilet base are coupled together; and a quick release coupling mechanism having a clip receiving portion and a clip configured to slide within the clip receiving portion from a disengage position to an engage position to engage the pump inlet coupling portion and couple the toilet base and the pump/motor assembly together, and also configured to slide from the engage position to the disengage position to disengage the pump inlet coupling portion and decouple the pump/motor assembly from the toilet base.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims benefit to provisional patent application Ser. No. 62/902,580, filed 19 Sep. 2019, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to apparatus having a toilet coupled to a motor/pump assembly, including where the toilet is an electric marine toilet.
2. Brief Description of Related Art
(2) There is no product known to the inventors that allows a user to release the pump/motor assembly from a toilet base to enable fast access to service/maintain the toilet. In view of this, there is a need in the industry for a better way to release a motor/pump assembly from a toilet to enable fast access to service/maintain the toilet.
SUMMARY OF THE PRESENT INVENTION
(3) In summary, the present invention provides a toilet base assembly that allows a user to perform fast servicing without the use of tools. On all electric marine toilets known the inventors, a series of screws have to be removed to allow access to the pump housing and impeller to remove blockages or service the toilet. According to the present invention, the pump/motor assembly will be removed by releasing a clip mechanism that holds the pump/motor assembly in place when in operation. The clip mechanism may include a clip that is manually inserted and removed, as well as a spring loaded clip.
PARTICULAR EMBODIMENTS

(4) By way of example, and according to some embodiments, the present invention may include, or take the form of, apparatus featuring a toilet base, a pump/motor assembly and a quick release coupling mechanism.
(5) The toilet base has a wall forming a chamber for receiving waste, the wall having a toilet base coupling portion.
(6) The pump/motor assembly may include a pump inlet configured to receive the waste pumped from the chamber. The pump inlet may include a pump inlet coupling portion arranged inside the toilet base coupling portion when the pump/motor assembly and the toilet base are coupled together.
(7) The quick release coupling mechanism may include a clip receiving portion and a clip. The clip may be configured to slide within the clip receiving portion from a disengage position to an engage position to engage the pump inlet coupling portion and couple the toilet base and the pump/motor assembly together. The clip may also be configured to slide from the engage position to the disengage position to disengage the pump inlet coupling portion and to decouple the pump/motor assembly from the toilet base.
Other Features
(8) The apparatus may include one or more of the following features:
(9) The quick release coupling mechanism may be mounted on the toilet base, and/or may form part of the toilet base coupling portion. By way of further example, embodiments are envisioned, and the scope of the invention is intended to include, mounting or configuring the quick release coupling mechanism on the pump/motor assembly, e.g., and having a corresponding toilet base coupling portion arranged inside a corresponding pump inlet coupling portion.
(10) The clip receiving portion may include a slot configured or formed therein to receive the clip; and the clip may be configured to slide within the slot of the clip receiving portion.
(11) The clip receiving portion may include an opening configured or formed therein; the pump inlet coupling portion may include a rim configured to pass through the opening formed in the clip receiving portion and into the toilet base coupling portion of the wall of the toilet base. The clip may be configured to slide within the clip receiving portion from the disengage position to the engage position and engage the rim to couple together the toilet base and the pump/motor assembly.
(12) The quick release coupling mechanism may include a spring configured to expand when the clip slides from the disengage position to the engage position to push and hold the clip in the engage position, and also configured to compress when the clip slides from the engage position to the disengage position to urge the spring back towards the engage position when the clip is released.
(13) The clip may include a clip engaging portion configured to engage the rim when the clip slides from the disengage position to the engage position, and also configured to disengage the rim when the clip slides from the engage position to the disengage position.
(14) The clip may include a clip release portion having an opening configured to allow the pump inlet coupling portion to pass thru when the clip slides into the disengage position for releasing the pump/motor assembly from the toilet base, and also for positioning the pump inlet coupling portion for engagement.
(15) The pump/motor assembly may include a pump outlet having a pump outlet coupling portion configured or formed thereon to provide the waste from the pump/motor assembly. The apparatus may include a discharge fitting having a quick release discharge fitting coupling mechanism with a discharge fitting clip receiving portion and a discharge fitting clip. The discharge fitting clip may be configured to slide within the discharge fitting clip receiving portion from a discharge fitting disengage position to a discharge fitting engage position to engage the pump outlet coupling portion and couple the discharge fitting and the pump/motor assembly together. The discharge fitting clip may also be configured to slide from the discharge fitting engage position to the discharge fitting disengage position to disengage the pump outlet coupling portion and decouple the discharge fitting from the pump/motor assembly.
(16) The discharge fitting clip receiving portion may include a discharge fitting slot configured or formed therein to receive the discharge fitting clip; and the discharge fitting clip may be configured to slide within the discharge fitting slot.
(17) The discharge fitting clip receiving portion may include an opening configured or formed therein. The pump outlet coupling portion may include a pump outlet rim configured to pass through the opening formed in the discharge fitting clip receiving portion and into the discharge fitting. The discharge fitting clip may be configured to slide within the discharge fitting clip receiving portion from the discharge fitting disengage position to the discharge fitting engage position to engage the pump outlet rim to couple

together the pump/motor assembly and the discharge fitting.
(18) The quick release discharge fitting coupling mechanism may include a discharge fitting spring that is configured to expand when the discharge fitting clip slides from the discharge fitting disengage position to the discharge fitting engage position to push and hold the discharge fitting clip in the discharge fitting engage position. The discharge fitting spring may also be configured to compress when the discharge fitting clip slides from the discharge fitting engage position to the discharge fitting disengage position to urge the discharge fitting spring back towards the discharge fitting engage position. The discharge fitting clip may include a discharge fitting clip engaging portion configured to engage the pump outlet rim when the discharge fitting clip slides from the discharge fitting disengage position to the discharge fitting engage position, and also configured to disengage the pump outlet rim when the discharge fitting clip slides from the discharge fitting engage position to the discharge fitting disengage position. The discharge fitting clip may include a discharge fitting release portion having an opening configured to allow the pump outlet coupling portion to pass thru when the discharge fitting clip slides into the discharge fitting disengage position for releasing the discharge fitting from the pump/motor assembly, and for also positioning the pump outlet coupling portion for engagement.
(19) The apparatus may include an electric toilet, e.g., such as a marine electric toilet or an RV electric toilet.

Description

BRIEF DESCRIPTION OF THE DRAWING
(1) The drawing, which is not necessarily drawn to scale, includes the following Figures:
(2) FIG. **1** shows an exploded view of apparatus having a quick release pump/motor assembly connected to a toilet base that may form part of an electric marine toilet, e.g., having a clip for engaging/releasing the pump/motor assembly to/from the toilet base, and having a discharge fitting clip with a spring configuration for engaging/releasing a discharge fitting to/from the pump/motor assembly, according to some embodiments of the present invention.
(3) FIG. **2**A shows an exploded view of apparatus including a quick release pump/motor assembly connected to a toilet base that may form part of an electric marine toilet, e.g., having a clip for engaging/releasing a pump/motor assembly to/from the toilet base, and having a discharge fitting clip for engaging/releasing a discharge fitting to/from the pump/motor assembly, according to some embodiments of the present invention.
(4) FIG. **2**B shows a photograph of the quick release pump/motor assembly shown in FIG. **2**A as assembled for connecting to the toilet base, according to some embodiments of the present invention.
(5) FIG. **2**C**1** shows a perspective view of the apparatus shown in FIG. **2**A as assembled, e.g., having the clip pushed in for coupling the pump/motor assembly and the toilet base, and also having the discharge fitting clip pushed in for coupling the discharge fitting and the pump/motor assembly, according to some embodiments of the present invention.
(6) FIG. **2**C**2** is a cross-sectional view of a quick release coupling mechanism showing the clip engaging part of a pump inlet of the pump/motor assembly shown in FIG. **2**C**1** for connecting the pump/motor assembly to the toilet base, according to some embodiments of the present invention.
(7) FIG. **2**C**3** is a cross-sectional view of a quick release discharge fitting coupling mechanism showing the discharge fitting clip engaging part of a pump outlet of the pump/motor assembly shown in FIG. **2**C**1** for connecting the pump/motor assembly and the discharge fitting, according to some embodiments of the present invention.
(8) FIG. **2**D**1** shows a perspective view of the apparatus shown in FIG. **2**C**1**, e.g., having the clip pulled out for disengaging the clip from part of the pump inlet of the pump/motor assembly and releasing the pump/motor assembly from the toilet base, and also having the discharge fitting clip pulled out for disengaging the discharge fitting clip from part of the pump outlet of the pump/motor assembly and releasing the discharge fitting from the pump/motor assembly, according to some embodiments of the present invention.
(9) FIG. **2**D**2** shows an exploded view of the apparatus shown in FIG. **2**D**1**, e.g., having the pump/motor assembly pulled out from the toilet base, and also having the discharge fitting pulled out of the pump outlet

of the pump/motor assembly, according to some embodiments of the present invention.
(10) FIG. **2**E**1** shows a partial perspective view of the apparatus shown in FIG. **2**C**1**, e.g., having a discharge fitting configured to be rotated between 0°-180° in relation to, and about a longitudinal axis of, the pump inlet, to orient the discharge fitting in opposite directions as needed depending on the application, according to some embodiments of the present invention.
(11) FIG. **2**E**2** shows a partial perspective view of the apparatus shown in FIG. **2**C**1**, e.g., having the discharge fitting configured to be rotated 360° in relation to, and about a corresponding longitudinal axis of, the pump outlet, to swivel the discharge fitting as needed depending on the application, according to some embodiments of the present invention.
(12) FIG. **2**F shows a side view of the apparatus shown in FIG. **2**C**1**, according to some embodiments of the present invention.
(13) FIG. **2**G shows a corresponding side view of the apparatus shown in FIG. **2**C**1** from the opposite side, according to some embodiments of the present invention.
(14) FIG. **2**H shows a back view of the apparatus shown in FIG. **2**C**1**, according to some embodiments of the present invention.
(15) FIG. **2**I shows a front view of the apparatus shown in FIG. **2**C**1**, according to some embodiments of the present invention.
(16) FIG. **2**J shows a top view of the apparatus shown in FIG. **2**C**1**, according to some embodiments of the present invention.
(17) FIG. **2**K shows a bottom view of the apparatus shown in FIG. **2**C**1**, according to some embodiments of the present invention.
(18) FIG. **3**A shows a photograph of apparatus, including a quick release pump/motor assembly connected to a toilet base that may form part of an electric marine toilet, according to some embodiments of the present invention.
(19) FIG. **3**B**1** shows a perspective view of the apparatus shown in FIG. **3**A, e.g., having a clip and spring configuration for engaging/releasing part of a pump inlet of a pump/motor assembly for connecting the pump/motor assembly to/from the toilet base, and also having a discharge fitting clip and spring configuration for engaging/releasing part of a pump outlet for connecting the discharge fitting to/from the pump/motor assembly, according to some embodiments of the present invention.
(20) FIG. **3**B**2** is a cross-sectional view of the quick release coupling mechanism showing the clip engaging part of the pump inlet of the pump/motor assembly shown in FIG. **3**B**1** for coupling the pump/motor assembly and the toilet base, according to some embodiments of the present invention.
(21) FIG. **3**B**3** is a cross-sectional view of the quick release discharge fitting coupling mechanism showing the discharge fitting clip engaging part of the pump outlet of the pump/motor assembly shown in FIG. **3**B**1** for coupling the discharge fitting and pump/motor assembly, according to some embodiments of the present invention.
(22) FIG. **3**C**1** shows a perspective view of the apparatus shown in FIG. **3**A, e.g., having the clip pushed in for compressing the spring, disengaging the clip from part of the pump inlet of the pump/motor assembly, and releasing the pump/motor assembly from the toilet base, and also having the discharge fitting clip pushed in for compressing the discharge fitting spring, disengaging the discharge fitting clip from part of the pump outlet of the pump/motor assembly, and releasing the discharge fitting from part of the pump outlet of the pump/motor assembly, according to some embodiments of the present invention.
(23) FIG. **3**C**2** is a cross-sectional view of the quick release coupling mechanism showing the clip engaging part of the pump inlet of the pump/motor assembly shown in FIG. **3**C**1** for connecting the pump/motor assembly to the toilet base, according to some embodiments of the present invention.
(24) FIG. **3**C**3** is a cross-sectional view of the quick release discharge fitting coupling mechanism showing the discharge fitting clip engaging part of the pump outlet of the pump/motor assembly shown in FIG. **3**C**1** for coupling the pump/motor assembly to the discharge fitting, according to some embodiments of the present invention.
(25) FIG. **3**D**1** is a cross-sectional view showing part of the pump/motor assembly and the quick release coupling mechanism shown in FIG. **3**C**1**, e.g., having the clip pushed out by the spring for engaging part of the pump inlet of the pump/motor assembly and for coupling the pump/motor assembly and the toilet base, according to some embodiments of the present invention.
(26) FIG. **3**D**2** is a cross-sectional view showing part of the pump/motor assembly and the quick release

coupling mechanism shown in FIG. **3**C**1**, e.g., having the clip pushed in for compressing the spring, disengaging the clip from part of the pump inlet of the pump/motor assembly, and releasing the pump/motor assembly from the toilet base, according to some embodiments of the present invention. (In FIG. **3**D**2**, the seven (7) “phantom” dots shown in the clip indicate where the spring extended before being compressed, and do not indicate any connection between the clip and the pump/motor assembly spring.)

(27) FIG. **3**E**1** is a cross-sectional view of part of the pump/motor assembly and the quick release discharge fitting coupling mechanism shown in FIG. **3**C**1**, e.g., having the discharge fitting clip pushed out by the discharge fitting spring for engaging part of the pump outlet of the pump/motor assembly and for coupling the discharge fitting and the pump/motor assembly, according to some embodiments of the present invention.

(28) FIG. **3**E**2** is a cross-sectional view of part of the pump/motor assembly and the quick release discharge fitting coupling mechanism shown in FIG. **3**C**1**, e.g., having the discharge fitting clip pushed in for compressing the discharge fitting spring, disengaging the discharge fitting clip from part of the pump outlet of the pump/motor assembly, and releasing the discharge fitting from the pump/motor assembly, according to some embodiments of the present invention. (In FIG. **3**E**2**, the six (6) “phantom” dots shown in the discharge fitting clip indicate where the discharge fitting spring extended before being compressed, and do not indicate any connection between the discharge fitting clip and the discharge fitting spring.)

(29) FIG. **3**F shows an exploded view of the apparatus shown in FIG. **3**C**1**, e.g., having the pump inlet of the pump/motor assembly pulled out from the toilet base, and also having the discharge fitting pulled out of the pump outlet of the pump/motor assembly, according to some embodiments of the present invention.

(30) FIG. **3**G**1** shows a partial perspective view of the apparatus shown in FIG. **3**C**1**, e.g., having a discharge fitting portion of the quick release pump/motor assembly configured to be rotated between 0°-180° in relation to, and about a longitudinal axis of, the pump/motor assembly, to orient the discharge fitting in opposite directions as needed depending on the application, according to some embodiments of the present invention.

(31) FIG. **3**G**2** shows a partial perspective view of the apparatus shown in FIG. **3**C**1**, e.g., having the discharge fitting configured to be rotated 360° in relation to, and about a corresponding longitudinal axis of, the pump outlet, to swivel the discharge fitting as needed depending on the application, according to some embodiments of the present invention.

(32) FIG. **3**H shows a side view of the apparatus shown in FIG. **3**C**1**, according to some embodiments of the present invention.

(33) FIG. **3**I shows a corresponding side view of the apparatus shown in FIG. **3**C**1** from the opposite side, according to some embodiments of the present invention.

(34) FIG. **3**J shows a back view of the apparatus shown in FIG. **3**C**1**, according to some embodiments of the present invention.

(35) FIG. **3**K shows a front view of the apparatus shown in FIG. **3**C**1**, according to some embodiments of the present invention.

(36) FIG. **3**L shows a top view of the apparatus shown in FIG. **3**C**1**, according to some embodiments of the present invention.

(37) FIG. **3**M shows a bottom view of the apparatus shown in FIG. **3**C**1**, according to some embodiments of the present invention.

(38) Similar parts or components in Figures are labeled with similar reference numerals and labels for consistency. Every lead line and associated reference label for every element is not included in every Figure of the drawing to reduce clutter in the drawing as a whole.

DETAILED DESCRIPTION OF THE INVENTION

(39) By way of example, FIGS. **1**, **2**A thru **2**K and **3**A thru **3**M show different embodiments of the present invention, which may include, or take the form of, apparatus generally indicated as **10**, **10**′, **10**″ featuring a toilet base **12**, a pump/motor assembly **14** and a quick release coupling mechanism generally indicated as **20**. The apparatus **10** may include an electric toilet, e.g., such as a marine electric toilet or an RV electric toilet. The upper part of the electric toilet is not shown, does not form part of the underlying invention, and may include some combination of a toilet bowl, a toilet seat, a flushing mechanism, etc.

The Basic Invention

(40) The toilet base **12** may include a wall **12***a* forming a chamber **12***b* for receiving waste. The wall **12***a* may be configured with, or may include, a toilet base coupling portion **12***c*, e.g., consistent with that set

forth herein.

(41) The pump/motor assembly **14** may include a pump inlet generally indicated as **14***a* configured to receive the waste pumped from the chamber **12***b*. The pump inlet **14***a* may include a pump inlet coupling portion **14***b* (e.g., see FIG. **2**A) for arranging inside the toilet base coupling portion **12***c* when the pump/motor assembly **14** and the toilet base **12** are coupled together.

(42) The quick release coupling mechanism **20** may include a clip receiving portion **22** and a clip **24**. The clip **24** may be configured to slide within the clip receiving portion **22** from a disengage position (see FIG. **2**D**1**, **3**C**2**) to an engage position (see FIG. **2**C**2**, **3**B**2**) to engage the pump inlet coupling portion **14***b* and couple the toilet base **12** and the pump/motor assembly **14** together. The clip **24** may also be configured to slide from the engage position (see FIG. **2**C**2**, **3**B**2**) to the disengage position (see FIG. **2**D**1**, **3**C**2**) to disengage the pump inlet coupling portion **14***b* and decouple the pump/motor assembly **14** from the toilet base **12**.

(43) The quick release coupling mechanism **20** may be mounted on the toilet base **12**, and/or may include, or may form part of the toilet base coupling portion **12***c*, e.g. consistent with that disclosed herein.

(44) The clip receiving portion **22** may include a slot **22***a* (e.g., see FIG. **2**A) configured or formed therein to receive the clip **24**. The clip **24** may be configured to slide within the slot **22***a* of the clip receiving portion **22**.

(45) The clip receiving portion **22** may include an opening **22***b* configured or formed therein. The pump inlet coupling portion **14***b* may include a rim **14***b*′ configured and dimensioned to pass through the opening **22***b* formed in the clip receiving portion **22** and into the toilet base coupling portion **12***c* of the toilet base **12**. The clip **24** may be configured to slide within the clip receiving portion **22** from the disengage position (see FIG. **3**C**2**) to the engage position (see FIG. **2**C**2**, **3**B**2**) to engage the rim **14***b*′ to couple together the toilet base **12** and the pump/motor assembly **14**.

(46) The quick release coupling mechanism **20** may include a spring **26** (see **3**D**1**, **3**D**2**) configured to expand when the clip **24** slides from the disengage position (see FIG. **3**C**2**) to the engage position (see FIG. **3**B**2**) to push and hold the clip **24** in the engage position. The spring **26** (see **3**D**1**, **3**D**2**) may also be configured to compress when the clip **24** is pressed and slides from the engage position to the disengage position to urge the spring **26** back towards the engage position. The spring **26** may be configured between a top portion **24***a* of the clip **24** and an interior portion **28** of the quick release coupling mechanism **20**, e.g., as shown in FIGS. **3**D**1**, **3**D**2**. Circled portion D**1** show a lower portion of the clip **24** having a detente **24***b* for sliding and contacting a corresponding detent **28***a* of the interior portion **28** of the quick release coupling mechanism **20** for stopping and keeping the clip **24** within the slot **22***a* when the clip **24** is in the engage position. Circled portion D**2** show the detente **24***b* of the clip **24** separated from the corresponding detent **28***a* of the interior portion **28** when the clip **24** is in the disengage position. (It is important to note that in operation the clip **24** and detente **24***b* are configured to slide in relation to the corresponding detent **28***a* of the interior portion **28**, and are not connect together with the corresponding detent **28***a* of the interior portion **28**.)

(47) The clip **24** may include a clip engaging portion **24***c* (see FIG. **3**B**2**) configured to engage the rim **14***b*′ when the clip **24** slides from the disengage position (see FIG. **3**C**2**) to the engage position (see FIG. **3**B**2**), and also configured to disengage the rim **14***b*′ when the clip **24** slides from the engage position to the disengage position. (In FIG. **3**B**2**, a lower part (not shown) of the rim **14***b*′ is engaged and covered by part of the clip engaging portion **24***c* of the clip **24**.)

(48) The clip **24** may include a clip release portion **24***d* having an opening **24***d*′ configured to allow the rim **14***b*′ to pass thru when the clip **24** slides into the disengage position (see FIG. **3**C**2**) for releasing the pump/motor assembly **14** from the toilet base **12**, and also for positioning the pump inlet coupling portion **14***b* and rim **14***b*′ for engagement. (In FIG. **3**C**2**, the rim **14***b*′ is not engaged by any part of the clip—release portion **24***d* of the clip **24**.)

(49) The pump/motor assembly **14** may include a pump outlet **14***d* (see FIG. **1**) having a pump outlet coupling portion **14***e* (FIG. **3**F) configured or formed thereon to provide the waste from the pump/motor assembly **14**.

(50) The apparatus **10** may also include a discharge fitting **30** having a quick release discharge fitting coupling mechanism **32** with a discharge fitting clip receiving portion **34** (FIG. **2**D**1**) and a discharge fitting clip **36**, e.g., consistent with that shown in FIGS. **2**D**1**, **2**D**2**. The discharge fitting clip **36** may be configured to slide within the discharge fitting clip receiving portion **34** from a discharge fitting disengage

position (see FIG. **3**C**3**) to a discharge fitting engage position (see FIG. **2**C**3**, **3**B**3**) to engage the pump outlet coupling portion **14***e* and couple the discharge fitting **30** and the pump/motor assembly **14** together. The discharge fitting clip **36** may also be configured to slide from the discharge fitting engage position (see FIG. **2**C**2**, **3**B**3**) to the discharge fitting disengage position (see FIG. **3**C**3**) to disengage the pump outlet coupling portion **14***e* and decouple the discharge fitting **30** from the pump/motor assembly **14**.

(51) The discharge fitting clip receiving portion **34** may include a discharge fitting slot **34***a* (FIGS. **2**D**1**, **2**D**2**) configured or formed therein to receive the discharge fitting clip **36**. The discharge fitting clip **36** may be configured to slide within the discharge fitting slot **34***a*.

(52) The discharge fitting clip receiving portion **34** may include an opening **34***b* configured or formed therein. The pump outlet coupling portion **14***e* (see **3**F) may include a pump outlet rim **14***e*′ configured to pass through the opening **34***b* formed in the discharge fitting clip receiving portion **34** and into the discharge fitting **30**. The discharge fitting clip **30** may be configured to slide within the discharge fitting clip receiving portion **34** from the discharge fitting disengage position (see FIG. **3**C**3**) to the discharge fitting engage position (see FIG. **2**C**3**, **3**B**3**) to engage the pump outlet rim **14***e*′ of the pump outlet **14***d* to couple together the pump/motor assembly **14** and the discharge fitting **30**.

(53) The quick release discharge fitting coupling mechanism **32** may include a discharge fitting spring **38** (see FIG. **3**E**1**, **3**E**2**) that is configured to expand when the discharge fitting clip **36** is pressed and slides from the discharge fitting disengage position (see FIG. **3**C**3**) to the discharge fitting engage position (see FIG. **3**B**3**) to push and hold the discharge fitting clip **36** in the discharge fitting engage position. The discharge fitting spring **38** may also be configured to compress when the discharge fitting clip **36** slides from the discharge fitting engage position to the discharge fitting disengage position to urge the discharge fitting spring **38** back towards the discharge fitting engage position. Similar to the clip **24**, the spring **38** may be configured between a top portion **36***a* of the clip **36** and a top portion **40***a* of a discharge fitting housing **40**, e.g., as shown in FIGS. **3**E**1**, **3**E**2**. Circled portion E**1** shows the discharge fitting clip **36** having a detente **36***b* for engaging a corresponding detente **40***b* of the discharge fitting housing **40** for retaining the discharge fitting clip **36** within the discharge fitting slot **34***a*. Circled portion E**2** show the detente **40***b* separated from the corresponding detent **36***b* when the discharge fitting clip **36** is in the disengage position. In operation, the clip **36** and detente **36***b* are configured to slide in relation to the corresponding detent **40***b* and the discharge fitting housing **40**.

(54) The discharge fitting clip **36** may include a discharge fitting clip engaging portion **36**C (FIG. **3**B**3**) configured to engage the pump outlet rim **14***e*′ when the discharge fitting clip **36** slides from the discharge fitting disengage position (see FIG. **3**C**3**) to the discharge fitting engage position (see FIG. **3**B**3**), and also configured to disengage the pump outlet rim **14***e*′ when the discharge fitting clip **36** slides from the discharge fitting engage position to the discharge fitting disengage position.

(55) The discharge fitting clip **36** may include a discharge fitting release portion **36***d* (FIG. **3**C**3**) having an opening **36***d*′ configured to allow the pump outlet coupling portion **14***e* to pass thru when the discharge fitting clip **36** slides into the discharge fitting disengage position (see FIG. **3**C**3**) for releasing the discharge fitting **30** from the pump/motor assembly **14**, and for also positioning the pump outlet coupling portion **14***e* for engagement.

Other Components/Parts

(56) As one skilled in the art would appreciate, the drawing includes other components/parts that are not labeled or described to reduce clutter in the Figures, and do not form part of the underlying invention, e.g., including O-rings (FIG. **1**) for sealing the pump inlet **14***a* in relation to the toilet base coupling portion **12***c*; nuts and bolts for coupling different parts of pump, motor and the pump/motor assembly together; etc.

Possible Applications

(57) Possible applications may include marine electric toilets for vessel, or Recreational Vehicle (RV) electric toilets, etc.

The Scope of the Invention

(58) The embodiments shown and described in detail herein are provided by way of example only; and the scope of the invention is not intended to be limited to the particular configurations, dimensionalities, and/or design details of these parts or elements included herein. In other words, one skilled in the art would appreciate that design changes to these embodiments may be made and such that the resulting embodiments would be different than the embodiments disclosed herein, but would still be within the overall spirit of the present invention.

(59) It should be understood that, unless stated otherwise herein, any of the features, characteristics, alternatives or modifications described regarding a particular embodiment herein may also be applied, used, or incorporated with any other embodiment described herein.
(60) Although the invention has been described and illustrated with respect to exemplary embodiments thereof, the foregoing and various other additions and omissions may be made therein and thereto without departing from the spirit and scope of the present invention.

Claims

1. An electric toilet comprising: a toilet base having a wall forming a chamber for receiving waste, the wall having a toilet base coupling portion; a pump/motor assembly having a pump inlet with a longitudinal axis and configured to receive the waste pumped from the chamber, the pump inlet having a pump inlet coupling portion arranged inside the toilet base coupling portion when the pump/motor assembly and the toilet base are coupled together, the pump/motor assembly having a pump outlet configured to receive a discharge fitting and provide the waste from the pump/motor assembly; a quick release coupling mechanism having a clip receiving portion and a clip configured to slide within the clip receiving portion from a disengage position to an engage position to engage the pump inlet coupling portion and couple the toilet base and the pump/motor assembly together, configured to slide from the engage position to the disengage position to disengage the pump inlet coupling portion and decouple the pump/motor assembly from the toilet base, the clip configured to engage the clip receiving portion and allow the pump outlet and discharge fitting to rotate between 0°-180° in relation to and about the longitudinal axis of the pump inlet to orient the discharge fitting in opposite directions as needed depending on an application.
2. The electric toilet according to claim 1, wherein the quick release coupling mechanism is mounted on the toilet base, including mounted on the wall of the toilet base or to the toilet base coupling portion of the toilet base.
3. The electric toilet according to claim 1, wherein the clip receiving portion has a slot configured or formed therein to receive the clip; and the clip is configured to slide within the slot of the clip receiving portion.
4. The electric toilet according to claim 1, wherein the clip receiving portion has an opening configured or formed therein; the pump inlet coupling portion has a rim configured to pass through the opening formed in the clip receiving portion and into the toilet base coupling portion of the toilet base; and the clip is configured to slide within the clip receiving portion from the disengage position to the engage position to engage the rim to couple together the toilet base and the pump/motor assembly.
5. The electric toilet according to claim 1, wherein the quick release coupling mechanism comprises a spring configured to expand when the clip slides from the disengage position to the engage position to push and hold the clip in the engage position, and also configured to compress when the clip slides from the engage position to the disengage position to urge the spring towards the engage position.
6. The electric toilet according to claim 4, wherein the clip comprises a clip engaging portion configured to engage the rim when the clip slides from the disengage position to the engage position, and also configured to disengage the rim when the clip slides from the engage position to the disengage position.
7. The electric toilet according to claim 6, wherein the clip comprises a clip release portion having an opening configured to allow the pump inlet coupling portion to pass thru when the clip slides into the disengage position for releasing the pump/motor assembly from the toilet base, and also for positioning the pump inlet coupling portion for engagement.
8. The electric toilet according to claim 1, wherein the pump outlet has a corresponding longitudinal axis and includes a pump outlet coupling portion; and the discharge fitting includes a quick release discharge fitting coupling mechanism with a discharge fitting clip receiving portion and a discharge fitting clip, the discharge fitting clip configured to slide within the discharge fitting clip receiving portion from a discharge fitting disengage position to a discharge fitting engage position to engage the pump outlet coupling portion and couple the discharge fitting and the pump/motor assembly together, and also from the discharge fitting engage position to the discharge fitting disengage position to disengage the pump outlet coupling portion and decouple the discharge fitting from the pump/motor assembly, the discharge fitting clip configured to engage the discharge fitting clip receiving portion and allow the discharge fitting to rotate 360° in relation

to and about the corresponding longitudinal axis of the pump outlet to swivel the discharge fitting as needed depending on the application.

9. The electric toilet according to claim 8, wherein the discharge fitting clip receiving portion has a discharge fitting slot configured or formed therein to receive the discharge fitting clip; and the discharge fitting clip is configured to slide within the discharge fitting slot.

10. The electric toilet according to claim 8, wherein the discharge fitting clip receiving portion has an opening configured or formed therein; the pump outlet coupling portion includes a pump outlet rim configured to pass through the opening formed in the discharge fitting clip receiving portion and into the discharge fitting; and the discharge fitting clip is configured to slide within the discharge fitting clip receiving portion from the discharge fitting disengage position to the discharge fitting engage position to engage the pump outlet rim to couple together the pump/motor assembly and the discharge fitting.

11. The electric toilet according to claim 8, wherein the quick release discharge fitting coupling mechanism comprises a discharge fitting spring that is configured to expand when the discharge fitting clip slides from the discharge fitting disengage position to the discharge fitting engage position to push and hold the discharge fitting clip in the discharge fitting engage position, and also configured to compress when the discharge fitting clip slides from the discharge fitting engage position to the discharge fitting disengage position to urge the discharge fitting spring towards the discharge fitting engage position.

12. The electric toilet according to claim 10, wherein the discharge fitting clip comprises a discharge fitting clip engaging portion configured to engage the pump outlet rim when the discharge fitting clip slides from the discharge fitting disengage position to the discharge fitting engage position, and also configured to disengage the pump outlet rim when the discharge fitting clip slides from the discharge fitting engage position to the discharge fitting disengage position.

13. The electric toilet according to claim 8, wherein the discharge fitting clip comprises a discharge fitting release portion having an opening configured to allow the pump outlet coupling portion to pass thru when the discharge fitting clip slides into the discharge fitting disengage position for releasing the discharge fitting from the pump/motor assembly, and for also positioning the pump outlet coupling portion for engagement.

14. The electric toilet according to claim 7, wherein the electric toilet is a marine electric toilet or an RV electric toilet.

15. The electric toilet according to claim 1, wherein the electric toilet comprises one or more O-rings configured to seal the pump inlet coupling portion of the pump inlet in relation to the toilet base coupling portion of the toilet base.

16. The electric toilet according to claim 1, wherein the pump outlet has a corresponding longitudinal axis that is perpendicular to the longitudinal axis of the pump inlet.

17. The electric toilet according to claim 8, wherein the electric toilet comprises one or more O-rings configured to seal the pump outlet coupling portion and the quick release discharge fitting coupling mechanism.

18. The electric toilet according to claim 8, wherein the corresponding longitudinal axis of the pump outlet is perpendicular to the longitudinal axis of the pump inlet.